

VK-RMD: Privacy-Friendly and Missing-Modality Robust Human Sensing for IoT-Enabled Intelligent Spaces

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Abstract—IoT-enabled intelligent spaces require continuous human sensing for smart homes, assistive monitoring, healthcare environments, and human–environment interaction. RGB cameras provide rich human appearance, contextual, and structural information, but long-term RGB deployment in private indoor spaces raises privacy, acceptance, and placement concerns. Non-RGB sensing modalities such as depth, LiDAR, mmWave, and WiFi-CSI are more suitable for privacy-sensitive IoT deployment, yet they are heterogeneous, noisy, and frequently unavailable because of occlusion, device failure, communication instability, or changing environmental conditions. This paper presents VK-RMD, a Visual-Keypoint Reliability-guided Missing-modality Distillation framework for privacy-friendly multimodal human perception in IoT-enabled intelligent spaces. The central idea is to use RGB only as training-time privileged information, compress its human-structure knowledge through a visual-keypoint representation, and train a deployable non-RGB multimodal Student that supports arbitrary available sensor combinations. VK-RMD combines random missing-modality training, reliability-guided multimodal fusion, and teacher-student distillation to jointly support human pose estimation and human activity recognition. On MMFi, VK-RMD achieves 75.18 mm average MPJPE over 31 HPE modality combinations and 85.15% average HAR accuracy over 15 modality combinations. Its non-visual Student further obtains 84.68 mm average MPJPE and 80.62% average HAR accuracy without RGB at inference. These results indicate that VK-RMD provides a practical direction for privacy-friendly and sensor-availability-aware human sensing in IoT intelligent spaces.

Index Terms—Internet of Things, intelligent spaces, human sensing, multimodal sensing, privacy-friendly perception, visual keypoints, missing modality, reliability-guided fusion, human pose estimation, human activity recognition.

I. Introduction

Human-centric Internet of Things (IoT) spaces are moving from isolated sensing devices toward continuous perception systems that understand human body structure, behavior, and interaction with the surrounding environment. Smart homes, assisted-living rooms, healthcare monitoring spaces, and embodied-intelligence platforms all require reliable human perception under long-term deployment. In such environments, human pose estimation (HPE) provides fine-grained spatial structure of the human body, while human activity recognition

(HAR) provides semantic understanding of human behavior. These two tasks are complementary: HPE evaluates whether a system can recover body geometry, whereas HAR evaluates whether the sensed evidence can support activity-level interpretation. A deployable IoT human sensing system should therefore handle both structural and semantic perception.

RGB cameras are powerful for HPE and HAR because they capture dense appearance, body configuration, objects, and contextual cues. However, camera-centric IoT sensing is difficult to deploy in private spaces. Raw RGB streams expose faces, clothes, home layout, and daily routines. In addition, illumination changes, occlusion, camera placement, and user acceptance can make long-term camera sensing unreliable. These issues do not mean that RGB information is useless; rather, they suggest that RGB should be used carefully. In this work, RGB is treated as training-time privileged information instead of an inference-time sensing modality.

Non-RGB sensors provide a more acceptable sensing basis for privacy-sensitive intelligent spaces. Depth sensors capture geometry, LiDAR provides sparse spatial point observations, mmWave radar captures motion and reflection patterns, and WiFi-CSI reflects human-induced channel variations. These modalities avoid direct visual appearance in different ways and are therefore attractive for IoT deployment. However, non-RGB signals are also weaker and more heterogeneous than RGB. Some modalities can be sparse, noisy, or absent because of sensor dropout, occlusion, device placement, communication failure, or environmental interference. A model that assumes a fixed and complete sensor set may perform well in a controlled laboratory protocol but degrade under realistic missing-modality conditions.

Prior multimodal human sensing and X-Fi/VK-style methods have demonstrated the feasibility of using keypoint-level or non-RGB representations for human perception. Missing-modality learning and teacher-student distillation also provide useful technical tools. Nevertheless, these ideas need to be integrated around the IoT deployment problem: how can a system exploit RGB knowledge during training, avoid raw RGB at inference, adapt to arbitrary sensor availability, and support both HPE and HAR? A simple collection of modalities is not enough, because real IoT spaces require privacy-aware input selection, missing-sensor robustness, and dynamic

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reliability estimation.

This paper proposes VK-RMD, a Visual-Keypoint Reliability-guided Missing-modality Distillation framework. Following the central storyline of this work, RGB provides training-time privileged knowledge, visual keypoints (VK) act as a privacy-friendly structured bridge, and the final Student is a non-RGB multimodal model for deployment. The Student is trained with randomly sampled modality subsets, learns from teacher supervision, and dynamically fuses modality-wise predictions according to estimated reliability. At inference, raw RGB is not required. The same framework is evaluated on HPE and HAR to demonstrate both human-structure perception and activity-level semantic understanding.

The contributions of this paper are summarized as follows:

- We formulate VK as a privacy-friendly structured bridge that transfers training-time RGB knowledge to deployable non-RGB multimodal human sensing in IoT-enabled intelligent spaces.
- We propose a missing-modality robust Student that supports arbitrary available sensor combinations through random missing-modality training and reliability-guided fusion.
- We validate the framework on both HPE and HAR using MMFi, with random split, cross-scene, and cross-subject protocols, and compare against reproduced X-Fi/VK baselines and ablation variants.

II. Related Work

A. IoT-Enabled Intelligent Spaces and Human Perception

IoT-enabled intelligent spaces combine sensing, communication, edge intelligence, and context-aware decision making. Human sensing is a central function in such systems because human status and behavior determine how the environment should respond. In smart homes and healthcare spaces, pose and activity information can support fall detection, rehabilitation monitoring, daily activity analysis, and human-robot interaction. Compared with conventional computer-vision benchmarks, IoT sensing systems must account for privacy, deployment cost, sensor dropout, edge computation, and long-term reliability.

Multimodal sensing has become important because no single modality is consistently reliable across all indoor conditions. RGB provides dense visual evidence but raises privacy concerns. Depth captures body geometry but is sensitive to range and occlusion. LiDAR provides spatial point measurements but may be sparse for human body details. mmWave radar can operate under weak lighting and partial occlusion, but its point clouds may be noisy. WiFi-CSI is infrastructure-friendly but indirect and environment-dependent. The MMFi dataset provides a useful benchmark for studying such heterogeneous sensing conditions. [TO BE FILLED: exact MMFi citation]

B. Privacy-Friendly Human Sensing and Visual Keypoints

Privacy-friendly sensing aims to reduce exposure of personally identifiable visual content while preserving task-relevant information. Non-RGB sensing is one solution, but purely non-RGB data may lack detailed human structure. VK or skeleton-like representations provide a middle ground: they preserve body geometry and motion traces while suppressing texture, clothing, face, and background appearance. This paper uses VK as a structured bridge rather than as a formal privacy guarantee. Keypoints can still reveal behavior patterns, and this limitation should be acknowledged in privacy-sensitive deployment.

The key difference between VK-RMD and a simple RGB replacement is the role assigned to VK. VK is not merely another input stream. It is used as an intermediate supervision signal that compresses RGB-derived body structure into a form that can be transferred to non-RGB modalities. This distinction matters for IoT systems because the final deployed model should not require raw RGB, while still benefiting from the structural knowledge that RGB can provide during training.

C. Multimodal Sensor Fusion for HPE and HAR

Multimodal HPE and HAR benefit from complementary sensor cues. In HPE, geometric information from depth or LiDAR can complement motion-sensitive signals from radar and WiFi. In HAR, temporal and motion patterns can be inferred from multiple sensing streams. Early fusion, late fusion, attention-based fusion, and transformer-based fusion have been widely explored. However, many fusion systems implicitly assume that all modalities are available and reasonably reliable.

IoT deployment violates this assumption. A sensor may be physically absent, temporarily unavailable, or degraded by environment-specific noise. Therefore, a practical fusion system should adapt to the current modality set and the reliability of each modality. VK-RMD follows this deployment view by learning modality-wise predictions and fusing them with estimated reliability weights.

D. Missing-Modality Learning in IoT Sensing Systems

Missing-modality learning has been studied through modality dropout, feature reconstruction, hallucination, shared latent spaces, and robust fusion. This paper does not claim that missing-modality learning itself is new. Instead, the contribution lies in connecting missing-modality training with privacy-friendly non-RGB human sensing in IoT spaces. Random missing-modality training simulates practical sensor dropout, while all-combination evaluation measures whether the trained Student remains usable under arbitrary sensor availability.

The missing-modality problem is especially important for HPE and HAR because the importance of each sensor changes with the task. HPE is sensitive to spatial structure, while HAR can often tolerate coarser motion cues. This difference motivates evaluating both tasks instead of relying on a single benchmark.

IoT intelligent space with RGB used only during training; deployable non-RGB sensors include depth, LiDAR, mmWave, and WiFi-CSI. The system outputs HPE and HAR while handling missing sensors.

Fig. 1. Overall IoT-enabled intelligent-space sensing scenario. RGB contains rich visual information but is privacy-sensitive. VK-RMD uses RGB only during training and deploys a non-RGB Student for HPE and HAR under missing sensor modalities.

E. Knowledge Distillation for Robust Sensor-Based Perception

Teacher-student distillation allows a deployable model to learn from stronger training-time information. Learning using privileged information and knowledge distillation are closely related to this design. In VK-RMD, the teacher can access richer structural cues during training, while the Student is constrained to use deployable modalities. This is suitable for IoT settings, where training may use more information than what is acceptable or available during deployment.

The paper does not claim that RGB-to-pose distillation is new. The distinction is that RGB knowledge is compressed through VK and then used to train a non-RGB multimodal Student under missing-modality conditions. Thus, distillation serves the deployment chain from RGB privileged information to privacy-friendly non-RGB inference.

III. System Overview and Problem Formulation

A. IoT-Enabled Intelligent Space Scenario

We consider an intelligent indoor space equipped with heterogeneous sensors. During data collection and training, RGB may be available, together with VK annotations or extracted visual keypoints, depth, LiDAR, mmWave, and WiFi-CSI. During deployment, raw RGB is not required because it exposes appearance-level private information. The deployed system receives only the available non-RGB sensors and predicts HPE and HAR outputs.

Fig. ?? illustrates the target scenario. The sensing system should support multiple sensor configurations rather than a fixed modality set. For example, a smart home may contain a depth sensor in one room, a WiFi router in another, and an mmWave device near a doorway. The model should not collapse when one sensor is absent.

B. Sensor Modalities and Privacy Constraints

Let X^{rgb} denote RGB data available only at training time, and let X^{vk} denote the visual-keypoint representation. The non-RGB modalities are X^d , X^l , X^m , and X^w , corresponding to depth, LiDAR, mmWave, and WiFi-CSI. We use $\mathcal{M}$ to denote the set of available modalities at inference. The deployed Student should operate for any non-empty $\mathcal{M}$ supported by the training setting.

Training: RGB $\rightarrow$ VK bridge $\rightarrow$ teacher supervision.
Student: non-RGB encoders $\rightarrow$ missing-modality mask $\rightarrow$ reliability-guided fusion $\rightarrow$ HPE/HAR heads.

Fig. 2. Overall framework of VK-RMD. RGB is used only as training-time privileged information. VK provides a privacy-friendly structural bridge. The deployed Student receives available non-RGB modalities, estimates modality reliability, and predicts HPE and HAR.

The privacy constraint is practical rather than formal. We avoid raw RGB at inference and use VK as a reduced appearance representation during training. VK preserves body structure but removes most visual texture and background details. It does not guarantee anonymity, because behavior and motion patterns may still be sensitive.

C. Missing-Modality Problem Definition

In realistic IoT spaces, the available modality set changes. A sensor can be missing because of hardware failure, network dropout, occlusion, or environmental noise. We model this by sampling modality subsets during training and evaluating all modality combinations during testing. The objective is not to force all combinations to perform equally, but to avoid catastrophic failure and provide stable predictions whenever enough sensing evidence is available.

D. HPE and HAR Task Formulation

For HPE, the target is a set of 3-D joint coordinates $Y^{hpe} \in \mathbb{R}^{J \times 3}$. We evaluate HPE using MPJPE and PA-MPJPE. For HAR, the target is an action class Y^{har} . We evaluate HAR using accuracy and macro-F1. HPE and HAR share the same sensing backbone but use different heads.

IV. Proposed Method

A. Overview of VK-RMD

VK-RMD consists of four components: VK-based structural supervision, multi-sensor feature encoding, random missing-modality training, and reliability-guided fusion. A teacher-student distillation mechanism connects training-time privileged information to deployable non-RGB sensing. Fig. ?? summarizes the framework.

B. Privacy-Friendly VK Representation

The VK representation acts as a structured bridge between RGB and non-RGB sensing. Compared with raw RGB, VK removes most appearance information. Compared with purely non-RGB signals, VK provides explicit human-structure information. During training, this bridge helps the Student learn a body-centric representation. During inference, VK may be included when a privacy-friendly keypoint source is available, but the non-visual Student can also operate without VK.

Heatmap placeholder: rows are modality combinations; columns are VK, depth, LiDAR, mmWave, and WiFi-CSI; values are learned reliability weights.

Fig. 3. Reliability-guided fusion visualization. The final submission should visualize average reliability weights for different modality combinations. The current draft marks this as a required figure to be filled from model outputs.

C. Multi-Sensor Feature Encoding

Each modality is processed by a modality-specific encoder and projected into a common representation space. The encoders account for the different data structures of depth images, LiDAR points, mmWave observations, WiFi-CSI, and VK coordinates. The projected tokens are then consumed by a fusion module. This design allows the model to use heterogeneous sensors without forcing them into a single raw format.

D. Random Missing-Modality Training

During training, modality subsets are randomly sampled. If $\mathcal{A}$ is the full training modality set, the Student receives a subset $\mathcal{M} \subseteq \mathcal{A}$. The subset sampling simulates sensor dropout and prevents the model from over-relying on a single modality. The same model can then be evaluated under all modality combinations.

E. Reliability-Guided Multimodal Fusion

For each available modality $i \in \mathcal{M}$, the model predicts a candidate output $\hat{Y}_i$ and a reliability weight α_i . The fused output is

$$\hat{Y} = \sum_{i \in \mathcal{M}} \alpha_i \hat{Y}_i, \quad \sum_{i \in \mathcal{M}} \alpha_i = 1, \quad \alpha_i \geq 0. \quad (1)$$

This formulation is used for both HPE and HAR, with task-specific heads. It differs from uniform averaging by allowing the system to assign more influence to more reliable modalities.

F. Teacher-Student Missing-Modality Distillation

The teacher uses richer training-time information and provides supervision to the Student. The Student is trained with task loss and distillation loss:

$$\mathcal{L} = \mathcal{L}_{task} + \lambda_{kd} \mathcal{L}_{kd} + \lambda_{rel} \mathcal{L}_{rel}. \quad (2)$$

For HPE, $\mathcal{L}_{task}$ is a pose regression loss. For HAR, it is a classification loss. The distillation term encourages the Student to approximate teacher predictions or representations under missing modalities. The reliability term regularizes adaptive fusion. The exact loss weights should be filled from the final configuration files.

G. HPE and HAR Prediction Heads

The HPE head outputs joint coordinates, and the HAR head outputs class logits. HPE evaluates whether the framework recovers fine-grained spatial structure. HAR evaluates whether the sensing representation supports behavior semantics. Their joint evaluation strengthens the IoT interpretation of the framework.

V. Experiments

A. Dataset and Experimental Protocol

Experiments are conducted on MMFi, a multimodal human sensing dataset containing synchronized sensing modalities. The HPE experiments use VK, depth, LiDAR, mmWave, and WiFi-CSI. The HAR experiments use VK, depth, LiDAR, and mmWave in the current implementation. We report random split, cross-scene, and cross-subject results when available. Detailed dataset statistics and subject/action/environment splits should be filled from the final MMFi protocol description.

B. Implementation Details

The HPE Student-VK model has 8.90M parameters and runs at 62.01 FPS in the measured setting. The HPE Student-NV model has the same parameter count and runs at 63.37 FPS. The reproduced full baseline has 5.74M parameters and 52.04 FPS. Peak memory is approximately 4.95 GB for the HPE models. HAR models contain 8.61M parameters in the available result files; FPS and memory are not yet reported for HAR.

C. Baselines and Reproduced X-Fi/VK Results

The reproduced X-Fi/VK results are treated strictly as baselines. They are not part of the proposed method. For HPE, the reproduced baseline obtains 244.91 mm average MPJPE over 31 combinations and 244.70 mm in the full-modality setting. For HAR, the reproduced baseline obtains 69.63% average accuracy over 15 combinations and 77.70% full-modality accuracy. These baselines provide a reference for evaluating missing-modality robustness and deployability.

D. Main Results on HPE

Table ?? reports the HPE results. VK-RMD Student-VK achieves 75.18 mm average MPJPE over 31 modality combinations and 50.17 mm in the full-modality setting. The non-visual Student-NV achieves 84.68 mm average MPJPE over 15 combinations and 51.03 mm in the full non-visual setting. These results are substantially better than the reproduced X-Fi/VK baseline under all-combination evaluation.

TABLE I

Comparison with related categories of work. Checkmarks indicate the capability emphasized by each category. Specific prior-work citations should be verified before submission.

Category	Non-RGB inference	Missing modalities	Reliability fusion	HPE	HAR	IoT deployment
RGB-based HPE/HAR	No	Limited	Limited	Yes	Yes	Limited
Single non-RGB sensing	Yes	No	No	Partial	Partial	Yes
Generic multimodal fusion	Partial	Limited	Partial	Yes	Yes	Partial
Missing-modality learning	Partial	Yes	Partial	Task-dependent	Task-dependent	Partial
Reproduced X-Fi/VK baseline	Yes	Limited	No	Yes	Yes	Partial
VK-RMD (ours)	Yes	Yes	Yes	Yes	Yes	Yes

TABLE II

Main HPE results on MMFi. Lower MPJPE is better. The reproduced X-Fi/VK result is used only as a baseline.

Method	Role	Protocol	Combinations	Avg. MPJPE (mm)	Full MPJPE (mm)
Reproduced X-Fi/VK baseline	Baseline	Random	31	244.91	244.70
VK-RMD Student-VK	Ours	Random	31	75.18	50.17
VK-RMD Student-NV	Ours	Random	15	84.68	51.03
VK-RMD Student-VK	Ours	Cross-scene	31	142.02	71.71
VK-RMD Student-NV	Ours	Cross-scene	15	153.89	78.96
VK-RMD Student-VK	Ours	Cross-subject	31	78.57	49.88
VK-RMD Student-NV	Ours	Cross-subject	15	90.74	52.45

Plot HPE MPJPE and HAR accuracy against the number of missing modalities using Table ??.

Fig. 4. Missing-modality robustness comparison. The final figure should plot the grouped statistics in Table ?? and compare random, cross-scene, and cross-subject protocols where space permits.

Teacher-student distillation diagram: full training information guides a deployable Student that receives incomplete non-RGB modalities.

Fig. 5. Teacher-student missing-modality distillation. The teacher uses training-time privileged information, while the deployed Student only consumes available privacy-friendly sensing modalities.

E. Main Results on HAR

Table ?? reports the HAR results. VK-RMD Student-VK achieves 85.15% average accuracy over 15 combinations and 96.56% full-modality accuracy. The non-visual Student-NV achieves 80.62% average accuracy and 96.12% full non-visual accuracy. Cross-scene and cross-subject results show that the method remains effective beyond the random split protocol.

F. Robustness Under Missing Modalities

Table ?? groups results by the number of missing modalities. For HPE Student-VK, MPJPE increases gradually from 50.17 mm with no missing modality to 124.73 mm when only one modality is available. For HAR Student-VK, accuracy decreases from 96.56% with all modalities to 70.00% when only one modality remains. This trend is expected in sensor-sparse IoT conditions and is more informative than reporting only full-modality performance.

G. Ablation Study

Table ?? summarizes the main ablation results. Uniform fusion performs worse than reliability-guided fusion in both tasks. For HPE, uniform fusion obtains 82.77 mm average MPJPE, while VK-RMD Student-VK obtains 75.18 mm. For HAR, uniform fusion obtains 67.46% average accuracy, while VK-RMD Student-VK obtains 85.15%. These results support the use of reliability-guided

fusion. The no-distillation HPE variant is strong, so the paper does not claim that distillation improves every setting. Instead, distillation is presented as part of the training-time privileged supervision chain, and stronger output-KD evidence should be added before making a broader claim.

H. Reliability Weight Analysis

The available result files support the benefit of reliability-guided fusion through the uniform-fusion ablation. However, the final submission should also include direct visualization of reliability weights. Such a figure should aggregate weights by modality combination and show whether the model assigns larger weights to more informative sensors. At the current stage, the figure is specified in Fig. ?? but its numerical heatmap should be filled from model outputs.

I. Complexity and IoT Deployment Analysis

Table ?? reports available complexity measurements. The HPE Student-VK and Student-NV models have 8.90M parameters and run above 62 FPS in the measured setting, suggesting feasibility for real-time or near-real-time intelligent-space sensing. FLOPs and HAR runtime should be measured before submission.

TABLE III

Main HAR results on MMFi. Higher accuracy is better. The reproduced X-Fi/VK result is used only as a baseline.

Method	Role	Protocol	Combinations	Avg. Acc. (%)	Full Acc. (%)
Reproduced X-Fi/VK baseline	Baseline	Random	15	69.63	77.70
VK-RMD Student-VK	Ours	Random	15	85.15	96.56
VK-RMD Student-NV	Ours	Random	7	80.62	96.12
VK-RMD Student-VK	Ours	Cross-scene	15	80.61	95.25
VK-RMD Student-NV	Ours	Cross-scene	7	76.70	95.41
VK-RMD Student-VK	Ours	Cross-subject	15	85.30	96.12
VK-RMD Student-NV	Ours	Cross-subject	7	81.26	95.70

TABLE IV

Grouped missing-modality robustness. HPE values are MPJPE in mm; HAR values are accuracy in percent.

Task and model	Missing 0	Missing 1	Missing 2	Missing 3	Missing 4
HPE Student-VK	50.17	53.54	60.78	78.11	124.73
HPE Student-NV	51.03	61.32	79.43	124.32	—
HAR Student-VK	96.56	93.83	87.55	70.00	—
HAR Student-NV	96.12	90.05	66.02	—	—

TABLE V

Ablation results. HPE uses average MPJPE in mm; HAR uses average accuracy in percent. Extreme encoder-ablation failure combinations are not expanded in the main paper.

Task	Setting	Avg. result	Full result	Interpretation
HPE	VK-RMD Student-VK	75.18	50.17	Main HPE model.
HPE	w/o distillation	72.68	48.83	Distillation claim must be cautious for HPE.
HPE	uniform fusion	82.77	52.28	Reliability-guided fusion improves robustness.
HAR	VK-RMD Student-VK	85.15	96.56	Main HAR model.
HAR	w/o distillation	82.86	95.99	Distillation helps HAR average robustness modestly.
HAR	uniform fusion	67.46	95.39	Reliability-guided fusion is important under missing modalities.

TABLE VI

Complexity and deployment-related measurements. Missing entries should be measured before submission.

Model	Params	FPS	Peak memory
HPE Student-VK	8.90M	62.01	4.95 GB
HPE Student-NV	8.90M	63.37	4.95 GB
HPE baseline	5.74M	52.04	4.94 GB
HAR Student-VK	8.61M	To be measured	To be measured

VI. Discussion

A. Why VK-RMD Fits IoT Intelligent Spaces

VK-RMD is designed for IoT deployment constraints rather than only benchmark performance. The framework avoids raw RGB at inference, supports heterogeneous non-RGB sensing, and evaluates arbitrary missing-modality combinations. These properties are important for intelligent spaces because installed sensors differ across rooms, users, and applications. A fixed-modality system may be impractical when one sensor is absent or unreliable.

B. Privacy Benefits and Remaining Risks

The method reduces appearance-level visual exposure by avoiding raw RGB during inference and using VK as a structured bridge. This is more privacy-friendly than camera-centric inference, but it is not a formal privacy

guarantee. Body motion, pose, and activity labels can still reveal sensitive behavioral patterns. Therefore, the paper should avoid overstating privacy and should use terms such as privacy-friendly or reduced visual exposure.

C. Limitations

Several limitations remain. First, the current reliability analysis needs direct weight visualization. Second, the HPE no-distillation ablation is competitive, so the distillation contribution should be stated carefully and strengthened with output-KD combination results. Third, the exploratory SuperTeacher results are not strong enough to be presented as a main contribution; they should remain in discussion or appendix. Fourth, real-world IoT deployment with live sensors remains future work.

D. Future Work

Future work should extend VK-RMD to streaming deployment, online sensor-quality estimation, edge-device acceleration, and stronger privacy analysis. Additional experiments should measure runtime on edge hardware and evaluate robustness under real sensor failures rather than only offline modality masking.

VII. Conclusion

This paper presented VK-RMD, a privacy-friendly and missing-modality robust multimodal human sensing framework for IoT-enabled intelligent spaces. RGB is used as training-time privileged information, VK acts as a structured bridge, and the deployed Student operates without raw RGB. Through random missing-modality training and reliability-guided fusion, VK-RMD supports HPE and HAR under arbitrary available sensor combinations. Experiments on MMFi show strong HPE and HAR performance under random, cross-scene, and cross-subject protocols. The results support VK-RMD as a practical direction for sensor-availability-aware human perception in privacy-sensitive IoT spaces.

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